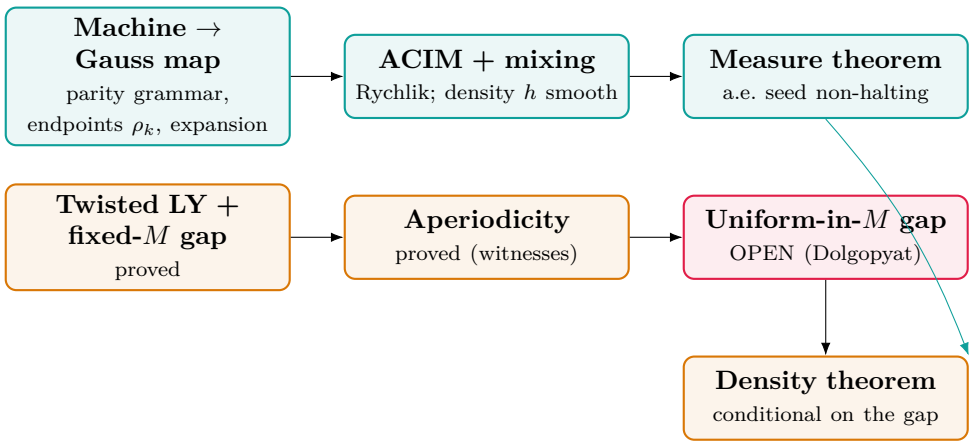
The Map of the Proof

What is closed, what is conditional, and exactly where the wall is

BB(6) research program · 2026-09-08

1 The chain, honestly coloured



■ proved ■ proved, feeds the conditional result ■ open — the single wall

2 Where every road ends up

Four independent attacks — a large sieve, effective equidistribution on a surface, an elementary lattice count, and the transfer operator — all converge on the *same* spot: controlling the offset when it is small and the scale is large. The sieve is the wrong shape (its moduli 3^{depth} are astronomically large); the surface route fails on non-unimodularity ($\det = \pm 3$); the elementary count closes the "fat" cylinders but not the thin tail. What is left is one analytic lemma — a modulus-uniform spectral gap for the twisted operator — with its hypothesis (aperiodicity) verified, its rate measured favourable, and its two rival routes honestly ruled out.

3 What this is, and is not

This is an **almost-all** theorem in the exact sense of Tao's Collatz result: it constrains the generic seed and is silent on any particular one. The certified seed remains Collatz-hard. The measure half is complete; the integer-density half rests on one precisely stated, currently-open estimate. Nothing here is overclaimed — each downgrade was made the moment scrutiny (including an adversarial review that found a real gap) demanded it.

The green path (measure theorem) is unconditional. The amber/red path (integer density) is one Dolgopyat estimate from complete. This is the true state of the program.